

Sara Oukajji

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Education

The University of Alabama,

Tuscaloosa, AL

Bachelor of Science in Mechanical Engineering

May 2028

GPA: 3.84/4.00

Technical Skills

- **CAD:** Siemens NX, SolidWorks, Creo, AutoCAD
- **Programming & Data:** Python, SQL, Power BI
- **Electrical:** Altium Designer, BMS, high-voltage
- **Languages:** English, Arabic, Berber
- **Certifications:** High Voltage, Microsoft Office Associate, L1 [High Power Rockets]

Experience

United Launch Alliance (ULA)

Decatur, AL

Transportation Engineering and Logistics Intern

May 2026 – Aug 2026

- Designing and prototyping a quad-cutter protective lid assembly for a new quad-cutter design.
- Developing a Ground Support Equipment (GSE) tracking checklist for Vulcan hardware to improve accountability and asset visibility.
- Creating Power BI dashboards and an automated shipping calendar to streamline logistics planning and shipment tracking.
- Supporting transportation and handling operations for Vulcan/Atlas launch vehicle components and associated support equipment.

National Aeronautics and Space Administration (NASA)

Eglin AFB, Stennis Space Center

Mechanical Design and Analysis Intern

Jan 2026 – May 2026

- Supported propulsion and engine test operations through the design and analysis of mechanical and piping systems for cryogenic infrastructure.
- Performed structural analysis for an \$8M infrastructure project, developing solutions to protect instrumentation from loads up to 40,000 lbf.
- Developed and organized product structures for test stand systems, improving model accessibility and engineering efficiency.

Innovative Combustion Technologies

Hoover, AL

Intern

May 2025 – Aug 2025

- Designed and analyzed combustion-related components to improve power system efficiency.
- Assisted in developing tools to predict SO_3 formation and support operational decision-making.
- Prepared technical documentation and reports for engineering reviews and research activities.

Extracurriculars/Professional Societies

Fuels & Combustion Lab

Tuscaloosa, AL

Research Assistant

Aug 2025 – May 2026

- Conducted research on aircraft contrail formation and atmospheric impacts, analyzing factors affecting condensation.
- Assisted in the design and development of a laboratory prototype to generate and study contrails under controlled conditions for experimental testing.

Formula SAE

Tuscaloosa, AL

Tractive System Lead

Aug 2024 – March 2026

- Helped launch the team's first-ever electric vehicle entry, taking on the challenge of starting from zero with no prior EV system in place.
- Integrated BMS and shutdown circuit, including precharge, contactors, IMD, and TSAL per FSAE rules.

Societies/Honors: UA Scholar Scholarship Recipient, Kroger Scholar, Honors College, Dean's List, Society of Women Engineers, American Society of Mechanical Engineers, Habitat for Humanity, Volleyball